11.3 Signal correlation support in commercial offerings

While it's still early days and not all correlation paths are supported by every provider, there is already core support for a number of correlations available in the commercial space. Oftentimes, these are related to traces–logs and metrics–traces support.

Lets first have a look at cloud providers and their native monitoring and observability solutions:

- AWS has a comprehensive correlation strategy based on OpenTelemetry (http://mng.bz/jPqV), with traces–logs correlation supported via AWS X-Ray and Amazon CloudWatch (http://mng.bz/WzWd).
- Like AWS, Azure is standardizing on OpenTelemetry. For example, see the Application Insights—based correlation support (http://mng.bz/8rAW).
- Google Cloud offers support for correlating log entries (https://cloud.google.com/logging/docs/view/correlate-logs) in its operations suite.

Now, we will move on to commercial offerings from observability ISVs. In the observability space, almost all vendors offer one or another form of correlation support in their solutions:

- AppDynamics has support for custom correlation in languages such as Java (http://mng.bz/EQvD) and .NET (http://mng.bz/N2px) for metrics to events.
- In Datadog, you can correlate metrics (https://docs.datadoghq.com/dashboards/correlations/), and they offer traces–logs correlation (https://www.datadoghq.com/blog/request-log-correlation/).
- With Dynatrace, you get a rich set of traces correlation (http://mng.bz/D4ng).
- Honeycomb provides a nice comparison (https://www.honeycomb.io/comparisons) of their approach to correlation to other APMs, pointing out things like their query speed and support for high-cardinality data analysis.
- Lightstep offers tracing-related (http://mng.bz/lWmR) correlation support.
- Logz.io offers traces–logs (http://mng.bz/Bm6l) correlation based on OpenSearch.
- In New Relic's offering, there is support for configuring correlation logic with decisions (http://mng.bz/d1R1), offering a blend of correlation paths.
- Splunk has an interesting approach, supporting the correlate command (http://mng.bz/rWXj) that lets you see an overview of the co-occurrence between fields in your telemetry data.
- Sumo Logic allows you to correlate signals in the context of security investigations via their Cloud SIEM (https://www.sumologic.com/blog/siem-investigation-correlation/) offering.

Now that you have an idea of both commercial offerings and open source tooling in the correlation space, let us move on to some general considerations on correlations and what to pay attention to when introducing them.